Theorem 0.1. (Goursat's Lemma)
Let f be holomorphic on an open set $G \subseteq \mathbb{C}$, and let $\triangle$ be a closed triangle in G. Let γ be the path along the edges of $\triangle$. Then

$$\int_\gamma f(z)dz = 0$$

Theorem 0.2.
Let f be a continuous function on a star-shaped region $G \subseteq \mathbb{C}$ such that

$$\int_\gamma f(z)dz = 0$$

for any triangle γ in G. Let a be such that $[a, z] \subseteq G$ for any $z \in G$. Then the function $F : G \to \mathbb{C}$ defined by:

$$F(z) = \int_{[a,z]} f(w)dw$$

is holomorphic in G, with $F' = f$

Theorem 0.3. (Cauchy's Integral Theorem for star-shaped regions)
Let G be a star-shaped region and let f be holomorphic on G. Then

$$\int_\gamma f(z)dz = 0$$

for every closed path γ in G

Theorem 0.4. (Cauchy's Integral Formula)
Let G be a region with f holomorphic on G. Let $z_0 \in G$ and let $\overline{D}(z_0; r) \subseteq G$. Let γ be the circle that forms the boundary of $\overline{D}(z_0; r)$ with the positive orientation. Then for any $z \in D(z_0; r)$,

$$f(z) = \frac{1}{2\pi i} \int_\gamma \frac{f(w)}{w - z} dw$$

Theorem 0.5.
Let $G \subseteq \mathbb{C}$ be an open set and $\triangle$ a closed triangle in G. Let γ denote the path along the edges of $\triangle$. Suppose that f is continuous on G and holomorphic on $G \setminus \{z_0\}$ where $z_0 \in \triangle$. Then

$$\int_\gamma f(z)dz = 0$$

Theorem 0.6.
Let $G \subseteq \mathbb{C}$ be a star-shaped region. Let γ be a closed path in G. Suppose that f is continuous on G and holomorphic on $G \setminus \{z_0\}$ where $z_0 \in G$. Then

$$\int_\gamma f(z)dz = 0$$

Theorem 0.7. (Cauchy's formula for derivatives)
Let G be a region, $z_0 \in G$ and $\overline{D}(z_0; r) \subseteq G$. Let γ be the circle that form the boundary of $\overline{D}(z_0; r)$, with positive orientation. Let f be a holomorphic function on G. Then $f^{(n)}(z)$ exists for any $z \in D(z_0; r)$ and

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_\gamma \frac{f(w)}{(w-z)^{n+1}} dw, \qquad n = 0, 1, 2, ...$$

Theorem 0.8.
If f is holomorphic on an open set G, then f has derivatives of all orders in G, and all derivatives are holomorphic in G

Theorem 0.9. (Taylor's Theorem)
Let f be a holomorphic function in $D(z_0; R)$ for some $R > 0$. Then there exists unique numbers $a_n \in \mathbb{C}, n = 0, 1, 2, ...$ such that

$$a_n = \frac{1}{2\pi i} \int_\gamma \frac{f(w)}{(w-z_0)^{n+1}} dw = \frac{f^{(n)}(z_0)}{n!}$$

Theorem 0.10. (Cauchy's Inequalities)
Assume that $|f(z)| \leq M$ for all z on the circle with center z_0 and radius r. Then the coefficients of Taylor's series $f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$ satisfy Cauchy's inequalities:

$$|a_n| \leq \frac{M}{r^n}, \quad n = 0, 1, 2, ...$$

Theorem 0.11.
Let f be an entire function. Assume there are constants $n \in \mathbb{N} \cup \{0\}, M, R > 0$ such that

$$|f(z)| \leq M|z|^n, \quad \text{for all } |z| \geq R$$

Then f is a polynomial of degree at most n

Theorem 0.12. (Liouville's Theorem)
An entire function that is bounded on $\mathbb{C}$ is constant

Theorem 0.13. (Fundamental Theorem of Algebra)
Let p be a non-constant polynomial with constant coefficients. Then p has a zero in $\mathbb{C}$

Theorem 0.14. Let $(f_n)_{n=1}^{\infty}$ be a sequence of functions holomorphic in a region $G \subseteq \mathbb{C}$ that converges to $f : G \to \mathbb{C}$, and the convergence is uniform on any compact subset of G. Then f is holomorphic in G, the sequence of derivatives $(f_n^{(k)})_{n=1}^{\infty}$ converges to $f^{(k)}$ for any $k \in \mathbb{N}$ and for every $z \in G$ there is a closed disc $\overline{D}(z;r) \subseteq G$ with some $r > 0$ such that the convergence is uniform in $\overline{D}(z;r) \subseteq G$

Theorem 0.15.
Assume that f is holomorphic in an open disc $D(z_0;r)$ with some $z_0 \in \mathbb{C}$ and $r > 0$. If z_0 is a limit point of the set $\{\xi \in D(z_0;r) : f(\xi) = 0\}$, then $f(z) = 0$ for all $z \in D(z_0;r)$

Theorem 0.16. (The Identity Theorem) Suppose that f is a holomorphic function in a region $G \subseteq \mathbb{C}$. If the set $\{\xi \in G : f(\xi) = 0\}$ has a limit point in G, then $f \equiv 0$ in G

Theorem 0.17.
If f is holomorphic in a region G and not constant, then the set $\{\xi \in G : f(\xi) = 0\}$ has no limit points in G

Theorem 0.18. (The Coincidence Principle)
Let f and g be holomorphic in a region G. If the set $\{\xi \in G : f(\xi) = g(\xi)\}$ has a limit point in G, then $f \equiv g$ in G

Theorem 0.19. (Gauss' Mean Value Theorem)
Let f be holomorphic in $D(z;R)$ for some $z \in \mathbb{C}$ and $R > 0$. Let $0 < r < R$. Then

$$f(z) = \frac{1}{2\pi}\int_0^{2\pi} f(z + re^{i\theta})d\theta$$

Theorem 0.20. (The Maximum Modulus Principle)
Let G be a bounded region and f holomorphic in G and continuous in $\overline{G}$. Then $|f|$ attains its maximum on the boundary of G. If f is non-constant, then

$$|f(z)| < \max_{w\in\partial G}|f(w)| \text{ for any } z \in G$$

Theorem 0.21. (Riemann's Extension Theorem)
If a function f is holomorphic and bounded in a punctured disc $D'(z_0;r)$ with some $z_0 \in \mathbb{C}$ and $r > 0$, then the limit $L = \lim_{z\to z_0} f(z)$ exists and the new function

$$\hat{f}(z) = \begin{cases} f(z) & z \in D'(z_0;r) \\ L & z = z_0 \end{cases}$$

is holomorphic in $D(z_0;r)$

Theorem 0.22. (The Casorati-Weierstrass Theorem)
Suppose thatt f is holomorphic in $D'(z_0; R)$ for some $R > 0$ and has an essential singularity at z_0. Then for any number $w_0 \in \mathbb{C}$, there exists a sequence $(z_n)_{n=1}^{\infty}$ such that $\lim_{n\to\infty} z_n = z_0$ and $\lim_{n\to\infty} f(z_n) = w_0$

Theorem 0.23.
Suppose that f is holomorphic in a punctured disc $D'(z_0; R)$ with some $R > 0$ and z_0 is a pole of f. Then there exists a natural number $n \in \mathbb{N}$ and a function h that is holomorphic in $D(z_0; r)$ with some $0 < r \leq R$ such that $h(z_0) \neq 0$ and

$$f(z) = \frac{1}{(z - z_0)^n h(z)}, \quad z \in D'(z_0; r)$$

Then the number n and the function h are unique

Theorem 0.24. (The Residue Theorem)
Let f be holomorphic in $\mathbb{C}$ except possibly for isolated singularities. Let γ be a positively oriented simple closed path in $\mathbb{C}$, and no singularities lie on γ. Let $z_1, z_2, ..., z_n$ be singularities of f that lie inside of γ. Then

$$\int_{\gamma} f(z)dz = 2\pi i \sum_{k=1}^{n} \text{Res}(f, z_k)$$

Theorem 0.25.
Let f be holomorphic in $D'(z_0; r)$ with some $r > 0$ and let z_0 be a pole of f.

(1) If z_0 is a simple pole of f, then $\text{Res}(f, z_0) = \lim_{z\to z_0}(z - z_0)f(z)$

(2) If $f(z) = \frac{g(z)}{h(z)}$ with $h(z_0) = 0, h'(z_0) \neq 0$ and $g(z_0) \neq 0$. Then $\text{Res}(f, z_0) = \frac{g(z_0)}{h'(z_0)}$

(3) If z_0 is a pole of order n, then $\text{Res}(f, z_0) = \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}}((z - z_0)^n f(z))|_{z=z_0}$

Theorem 0.26.
Let f be a function holomorphic in $D'(\infty, R)$ with some $R > 0$. Then

$$\text{Res}(f, \infty) = -\text{Res}\left(\frac{\hat{f}(z)}{z^2}, 0\right)$$